Difference-in-Differences With Multiple Time Periods

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1. Introduction

The Difference-in-Differences (DiD) method is one of the most widely used tools in applied economics and related fields for evaluating the causal effects of policies and interventions (Currie et al., 2020; Roth et al., 2023). The appeal of the method lies in its simplicity: by comparing the evolution of outcomes for treated and untreated groups over time, DiD controls for unobserved time-invariant differences across groups as well as for common shocks. In its classical form with two groups and two time periods, the design is intuitive, and its key identifying assumption, the parallel trends assumption, is straightforward to communicate.

In practice, however, empirical research often involves richer settings. Data typically span many periods, and treatment is introduced at different times across groups. Minimum wage reforms, education policies, or environmental regulations, for example, are rarely implemented simultaneously for all units. For a long time, two-way fixed effects (TWFE) regressions were the standard way of extending the basic DiD design to these cases, producing a single coefficient that summarized the treatment effect.

Recent research has shown that TWFE is not always reliable in these multi-period, staggered adoption settings (Baker et al., 2022; Goodman-Bacon, 2021). When treatment effects vary across groups or evolve dynamically, the TWFE estimator may not recover a well-defined average treatment effect on treated (ATT). Instead, it combines information from multiple two-by-two comparisons, some of which are invalid because already-treated groups are used as controls for not-yet-treated groups. This mixing of comparisons leads to biased estimates, and in some cases the underlying weighting scheme places negative weights on certain contrasts, so that the overall estimate can even fall outside the range of true effects. These insights have generated a rethinking of how DiD should be implemented when adoption is staggered.

The aim of this thesis is to study these issues in a controlled simulation environment. Using simulated data has the advantage that the true treatment effects are known by construction, which makes it possible to evaluate directly the performance of different estimators. A sequence of data-generating processes is used to show when TWFE performs well and when it fails. The most complex scenario, featuring staggered adoption with dynamic and heterogeneous effects, then serves as a benchmark for comparing modern alternatives.

The thesis focuses on two recent approaches that have been proposed as solutions to the limitations of TWFE: Callaway & Sant'Anna (2021) and Sun & Abraham (2021). Both methods avoid the contamination of control groups that plagues TWFE. The former estimates group-time average treatment effects by comparing each treated cohort only with not-yet-treated units and then aggregates them transparently. The latter provides a correction for event-study designs, ensuring that dynamic treatment profiles are estimated solely from valid comparisons. By applying these estimators to the simulate data, the thesis shows how they succeed in situations where TWFE breaks down.

The contribution of the thesis is therefore twofold. First, it provides a clear, simulation-based illustration of the conditions under which TWFE can and cannot be used to identify treatment effects. Second, it evaluates the Callaway-Sant'Anna (2021) and Sun-Abraham (2021) methods in the most realistic simulation design, highlighting their advantages relative to the traditional estimator.

The remainder of the thesis is structured as follows. Chapter 2 reviews the relevant literature, beginning with the classical DiD framework, the two-way fixed effects estimator, and its decomposition as discussed by Goodman-Bacon (2021), before turning to recent methodological contributions. Chapter 3 presents the methodological framework. Chapter 4 describes the simulation design. Chapter 5 and Chapter 6 report the results of the simulation study. Chapter 7 discusses the findings and concludes.

2. Background and Literature Review

This Chapter reviews methodological developments in Difference-in-Differences. It begins with the classical two-period framework, the discusses the extension to multiple periods through the two-way fixed effects estimator, and finally introduces recent advances that address its shortcomings.

2.1 Classical Difference-in-Differences

The classical DiD framework considers two groups: treated and untreated and two time periods, before and after treatment. Outcomes are modelled as potential outcomes: each unit has an outcome if treated and an outcome if untreated, but only one is observed in practice. The treatment effect is the difference between these potential outcomes, and the DiD design recovers this effect by differencing out permanent group-specific differences and common time shocks.

Identification rests on (i) the parallel trends assumption, which requires that, in the absence of treatment, the treated group would have experienced the same change in outcomes as untreated group, and (ii) no participation assumption, ensuring no pre-treatment effects. Under these assumptions, the DiD estimator recovers the average treatment effect on the treated (ATT). This setup is straightforward and powerful in its simplicity, but it is restrictive, since it assumes only two periods and a uniform treatment date across all treated units.

2.2 The Extension to Multiple Periods

Most empirical settings involve more than two periods and frequently feature staggered treatment adoption, where different units adopt the treatment at different points in time. To accommodate this, researchers have relied on the two-way fixed effects (TWFE) model, which regresses outcomes on unit fixed effects, time fixed effects, and a treatment indicator. When treatment effects are homogeneous and the parallel trends

assumption holds across all groups and periods, the coefficient on the treatment indicator can be interpreted as the ATT.

However, as highlighted in more recent econometric research (Borusyak et al., 2024; De Chaisemartin & D'Haultfœuille, 2020), the TWFE model does not behave well when treatment effects vary across groups or over time. In these cases, the regression does not estimate a convex average of treatment effects but instead produces a weighted average that may even assign negative weights to certain comparisons. This raises significant concerns for interpretation and has motivated a growing literature reassessing the validity of TWFE in staggered adoption settings.

2.3 The Goodman-Bacon Decomposition

An important contribution to understanding the behavior of TWFE was provided by Goodman-Bacon (2021), who showed that the TWFE estimate in staggered adoption designs can be written as a weighted average of all possible two-by-two DiD comparisons. These comparisons involve treated groups against never-treated groups, earlier-treated groups against not-yet-treated groups, and later-treated groups against already-treated groups. The first two categories are conceptually valid, but the third category is more limited, since already-treated units may not provide a credible counterfactual for later-treated groups.

The decomposition further reveals that some of the weights may be negative. This occurs when already-treated units act as controls, which can push the TWFE estimate in the opposite direction of the true treatment effect. The decomposition therefore makes clear why TWFE may produce estimates outside the range of the actual group-specific effects and why it should be used with caution when treatment effects are heterogeneous.

2.4 Callaway and Sant'Anna (2021)

To address these limitations, Callaway & Sant'Anna (2021) propose an estimator that focuses directly on group-time average treatment effects. For each group defined by its adoption period, they estimate the effect of treatment in each period relative to not-yet-treated units. This ensures that each comparison relies only on valid control groups, under a parallel trends assumption applied locally to treated and not-yet-treated units within each cohort. Once the group-time effects are estimated, they can be aggregated in different ways to recover cohort-specific averages, time-specific averages, or an overall ATT.

The main advantage of this approach is that the aggregation weights are explicit and non-negative. This contrast with TWFE, when the weighting is implicit and may involve negative values. By making the weights transparent and interpretable, Callaway & Sant'Anna provide a flexible framework that accommodates heterogeneous and dynamic treatment effects while avoiding the pitfalls of TWFE.

2.5 Sun and Abraham (2021)

While Callaway & Sant'Anna (2021) focus on group-time average treatment effects on treated, Sun & Abraham (2021) propose a modification of the event-study framework commonly used in applied research. Standard TWFE event studies estimate the effect of treatment at different leads, and lags relative to adoption, but these coefficients are biased because already-treated units often serve as controls. Sun & Abraham correct for this by estimating cohort-specific event-study coefficients and then aggregating them with valid weights. Their estimator ensure that each event-time coefficient is identified from comparisons with not-yet-treated groups, and thus relies on a parallel trends assumption that applies only to these groups.

Although this adjustment reduces the effective sample size, often resulting in wider confidence intervals, it preserves the intuitive regression-based event-study format that

applied researchers frequently use. By correcting for contamination from already-tread units, the Sun-Abraham estimator delivers unbiased event-time coefficients that can be meaningfully interpreted.

Beside two contributions above, several other approaches also have extended DiD to settings with staggered treatment adoption in different ways, for instance, Borusyak et al., (2024) with imputation estimator, Cengiz et al., (2019) with stacked DiD.

It is important to note that the methods discussed in this thesis concentrates on the case of binary treatment, in contrast to some recent studies that extend DiD to continuous or multi-valued treatments (Callaway et al., 2024; De Chaisemartin et al., 2024).

The shortcomings of TWFE and the weighting problem clarified by Goodman-Bacon, (2021) have motivated the development of new estimators. Among them, the approaches of Callaway & Sant'Anna (2021) and Sun & Abraham (2021) stand out as robust alternatives for staggered adoption settings. The next chapter presents these methodological frameworks in detail.

3. Methodology

This chapter outlines the econometric methods used in the simulation study. It first presents the TWFE estimator and the Goodman-Bacon decomposition, which clarifies how TWFE combines different comparisons. It then introduces the Callaway & Sant'Anna (2021) and Sun & Abraham (2021) estimators as modern alternatives designed for settings with staggered adoption and heterogeneous treatment effects.

3.1 The Two-Way Fixed Effects (TWFE) Estimator

Consider panel data $\{Y_{it}, D_{it}\}$ for units $i = 1, \dots, N$ observed over periods $t = 1, \dots, T$ where $D_{it} = 1$ if unit i is treated at time t and zero otherwise. The conventional TWFE regression takes the form

$$Y_{it} = \alpha_i + \lambda_t + \delta D_{it} + \varepsilon_{it},$$

where α_i are unit fixed effects, λ_t are time fixed effects, and δ is the coefficient on the treatment indicator. When treatment effects are homogeneous across units and time, and the parallel trends assumption hold, δ identifies the average treatment effect on the treated (ATT).

In setting with multiple periods and staggered adoption, however, the interpretation of δ is more complicated. The estimator implicitly combines comparisons across all groups and periods, including situations where already-treated groups act as controls for later-treated ones. When treatment effects vary across cohorts or evolve dynamically over time, the TWFE coefficient no longer corresponds to a meaningful average of treatment effects, and may even be biased in sign (De Chaisemartin & D'Haultfœuille, 2020)

3.2 The Goodman-Bacon Decomposition

Goodman-Bacon (2021) shows that the TWFE estimator in settings with staggered treatment adoption can be written as a weighted average of all possible two-by-two DiD comparisons between groups. Some of these comparisons are valid, such as using not-yet-treated units as controls, while others are problematic because they involve already-treated units serving as controls.

A key insight is that the weights assigned to these comparisons can be negative when treatment effects are heterogeneous. This implies the overall TWFE estimate may not correspond to any meaningful ATT, and in extreme cases can even take the opposite sign of the true underlying effects.

The decomposition itself is not a new estimator but rather a diagnostic tool. It shows how TWFE aggregates comparisons information across cohorts and highlights why approaches such as Callaway & Sant'Anna (2021) and Sun & Abraham (2021) have become preferred tools for analyzing staggered adoption designs.

3.3 Callaway and Sant'Anna (2021)

Callaway & Sant'Anna (2021) propose an estimator for DiD in settings with multiple periods and staggered adoption of treatment. Their framework is built on the idea of group-time average treatment effects, which serve as the fundamental causal parameters.

Set up. Following the framework of Callaway & Sant'Anna (2021), suppose there are T periods, indexed by $t = 1, \dots, T$.

Let D_{it} be a binary indicator equal to 1 if unit i is treated in period t, and 0 otherwise. Define G_g as a binary variable equal to 1 if unit i first receives treatment in period g, and let C denote a binary variable equal to 1 if unit i is never treated. For each unit, exactly one of $\{G_1, \dots, G_T\}$ or C takes the value 1.

The generalized propensity score is given by

$$p_g(X) = P(G_g = 1 \mid X,\, G_g + C = 1),$$

which is the probability that a unit belongs to treatment group g, conditional on covariates X and on being either treated in some period or never treated.

For a cohort of units first treated in period g, the effect at time t is defined as

$$ATT(g,t) = \mathbb{E}[\, Y_t(1) - Y_t(0) \mid G_g = 1\,],$$

where $Y_t(1)$ and $Y_t(0)$ denote the potential outcomes with and without treatment.

Identification. Identification relies on two key assumptions. The first is Conditional parallel trends assumption. Callaway & Sant'Anna (2021) propose two variants: conditional parallel trends based on never-treated groups:

$$\mathbb{E}[\, Y_t(0) - Y_{g-1}(0) \mid G_g = 1, X\,] = \mathbb{E}[\, Y_t(0) - Y_{g-1}(0) \mid G_g = 0, X\,],$$

And conditional parallel trend based on not-yet-treated groups:

$$[\, Y_t(0) - Y_{t-1}(0) \mid X,\, G_g = 1\,] = \mathbb{E}[\, Y_t(0) - Y_{t-1}(0) \mid X,\, D_s = 0,\, G_g = 0\,]$$

for some $s \geq t$.

These conditions require that, in the absent of treatment, the evolution of outcomes for treated group would have mirrored that of either never-treated groups or not-yet-treated groups, after conditioning on covariates X. The second is No anticipation, which rules out changes in outcomes before treatment actually occurs

Callaway & Sant'Anna (2021) demonstrate that, given these assumptions, the group-time average treatment can be identified in a semi-parametric framework as:

$$ATT(g,t) \;=\; \mathbb{E}\left[\left(\frac{G_g}{\mathbb{E}[G_g]} - \frac{\frac{p_g(X)C}{1-p_g(X)}}{\mathbb{E}\left[\frac{p_g(X)C}{1-p_g(X)}\right]}\right)(Y_t - Y_{g-1})\right]$$

This intuition behind this equation is that it is a weighted mean of the long difference in outcomes, where the weights are derived from the propensity score and normalized to sum to one. The idea is to focus only on group g and the control group, and to reweight the control observations so that those with covariates similar to units in group g receive higher weight, while those with less comparable characteristics receive lower weight.

Since one typically estimates many group–time effects, the procedure also defines ways of aggregating them into interpretable summary measures. A general aggregation takes the form

$$\theta = \sum_{g,t} w\,(g,t)\; ATT(g,t),$$

where the weights $w(g,t)$ are chosen to reflect the research design. Common examples include:

the overall average treatment effect (weights proportional to group sizes), dynamic treatment effects by event time $e = t - g$, and calendar–time effects that summarize impacts across all treated groups in a given period.

Inference is carried out using a multiplier bootstrap procedure, which perturbs the influence functions associated with the estimators rather than re–estimating models in each replication. This method is computationally efficient and allows the construction

of simultaneous confidence bands for the full path of dynamic effects

3.4 Sun and Abraham (2021)

Sun & Abraham (2021) focus on the event-study specification, where treatment is modeled through a sequence of leads and lags rather than a single indicator. They demonstrate that, under staggered adoption, conventional lead/lag regressions can also produce estimates lacking a clear causal interpretation, similar to the issues in the binary treatment case. Their method addresses this by estimating dynamic effects separately for each treatment cohort (corresponding to group G_g in Callaway & Sant'Anna (2021)), and then aggregating these estimates through a weighted average, with weights given by each cohort's relative sample size. The resulting parameter of interest is the cohort-specific average treatment effect on the treated ($CATT$), defined k periods after initial treatment, which is conceptually parallel to $ATT(g,t)$ in Callaway & Sant'Anna (2021).

The key theoretical insight of the paper is that, even within an event-study specification rather than a single binary treatment indicator, the TWFE lead/lag coefficients may still be biased. This arises because the weights placed on different $CATTs$ are not guaranteed to be positive unless treatment effects are homogeneous. In particular, the fixed effects estimands for k periods relatives to treatment can be expressed as non-convex combinations of the $CATT$ for that period together with $CATTs$ from other periods.

Therefore, beside two main assumptions: Parallel trend in baseline outcome, which is similar to conditional parallel trend based on never-treated, and no anticipation, Identification of Sun and Abraham method relies on the third assumption: homogeneity of treatment effects across cohorts in some restrictive cases.

Sun & Abraham (2021) introduce an alternative estimation method that relies on a fully interacted regression, saturating the specification with both relative-time

indicators D_{it}^{k} and cohort dummies $\mathbf{1}\{G = g\}$. This setup yields a separate estimate of the cohort-specific treatment effect, $CATT_{g,k}$, for each group g at relative time k. They refer to this approach as the interaction-weighted (IW) estimator. Formally, the model is

$$y_{it} = \alpha_i + \lambda_t + \sum_{g} \sum_{k \neq -1} \delta_{g,k} \left(\mathbf{1}\{G = g\} \cdot D_{it}^{k} \right) + \varepsilon_{it}$$

In this framework, only never-treated units (or, if absent, the last-treated group, which serves only as a control) provide the comparison outcomes. Event-study coefficients for each relative period k are then constructed by averaging the cohort-specific effects across groups, with weights equal to the share of each cohort observed in that period

Inference. For inference, Sun and Abraham show that the IW estimator is consistent and asymptotically normal under the parallel trends and no-anticipation assumptions, allowing for conventional standard errors. Unlike Callaway and Sant'Anna (2021), they do not rely on bootstrap for uniform bands, but instead focus on pointwise confidence intervals for each relative period.

4. Simulation Designs.

4.1 Purpose of the Simulation

The purpose of the simulation study is to provide a controlled environment in which the behavior of different Difference-in-Differences estimators can be evaluated. Unlike empirical applications, where the true causal effect of treatment is unknown, simulations allow the researcher to specify data-generating processes (DGPs) in which the underlying treatment effects are transparent. This makes it possible to assess directly whether an estimator recovers the true effect, how much bias arises under certain conditions, and how precise the resulting estimates are.

In the context of this thesis, the simulations serve two distinct goals. The first is to demonstrate the well-documented shortcomings of the two-way fixed effects (TWFE) estimator when applied to settings with staggered treatment adoption and

heterogeneous effects. By generating outcome paths under alternative DGPs, it becomes clear how TWFE can mis-weight comparisons across groups and periods, sometimes producing estimates that diverge substantially from the true ATT. The second goal is to evaluate modern alternatives, in particular the estimators proposed by Callaway & Sant'Anna (2021) and Sun & Abraham (2021). These methods are designed to correct for the problems inherent in TWFE, and the simulation framework provides a means of assessing whether they succeed in doing so in finite samples.

The simulations therefore play a dual role within the thesis: they motivate the need to move beyond the traditional TWFE approach and provide evidence on the relative strengths and limitations of recent advance in the literature.

4.2 Description of Data-Generating Processes

To evaluate estimator performance, the simulations are based on a balanced panel data set with $T = 36$ periods (covering the years 1990 to 2025) and $N = 600$ observational units indexed by i. Each unit belongs to one of 30 randomly drawn states, which determine whether and when treatment is adopted.

The outcome variable Y_{it} combines unit-specific fixed effects, time-specific shocks, a stochastic error term, and a treatment effect component. The unit fixed effects and year effects are independently drawn from a normal distribution with mean and variance 0.5^2 . The idiosyncratic error term is also drawn from a normal distribution with mean zero and variance 0.5^2.

The first three data-generating processes (DGPs) are designed to illustrate conditions under which the TWFE estimator is expected to perform well. Each DGP builds progressively in complexity: beginning with a simple single-treatment timing and constant effect (DGP1), moving to dynamic treatment effects (DGP2), and then to staggered treatment adoption across cohorts (DGP3).

DGP 1: Single Treatment Period, Constant Treatment Effects

In the first design, half of the states initiate treatment in 2008, while the remaining half remain untreated throughout the sample. For treated states, the outcome variable for unit i in year t is given by

$$Y_{it} = \alpha_i + \lambda_t + \varepsilon_{it} + \tau_{it},$$

where α_i denotes the unit fixed effect, λ_t is the time fixed effect, $\varepsilon_{it} \sim N(0, 0.5^2)$ is an idiosyncratic error term, and the treatment effect τ_{it} is drawn from a normal distribution with mean

$$\mathbb{E}[\tau_{it}] = 2 \cdot 1\{t \geq 2008\} \cdot 1\{\text{treated}\},$$

and variance 0.2^2.

This specification implies that treated units experience a constant increase of 2 in the outcome variable following adoption in 2008, with small unit-level variation around this effect. Under this design, the TWFE estimator is expected to recover the true average treatment effect, since treatment effects are homogeneous and occur at a single point in time.

DGP 2: Single Treatment Period, Dynamic Treatment Effects

The second design also assumes that half of the states adopt treatment in 2008, with the remainder never treated. In this case, however, treatment effects evolve dynamically over time. The outcome variable is

$$Y_{it} = \alpha_i + \lambda_t + \varepsilon_{it} + \tau_{it},$$

with the treatment effect defined as

$$\tau_{it} = \delta_i \cdot (t - 2008) \cdot 1\{t \geq 2008\} \cdot 1\{\text{treated}\},$$

where $\delta_i \sim N(0.3,\ 0.2^2)$.

This specification implies that treated units experience incremental gains over time following adoption, with the slope of the gain varying across units. The average effect

across treated units increases by approximately 0.3 per year after 2008. The TWFE estimator remains valid under this setting, since treatment timing is homogeneous, though variation in dynamics across units introduces additional dispersion in realised outcomes.

DGP 3: Staggered Treatment Adoption, Constant Treatment Effects

The third design introduces staggered adoption. Units are again incorporated into 30 randomly drawn states, but these states are now evenly divided into three treatment groups based on adoption timing. Specifically, one third of the states adopt treatment in 1999, another third in 2008, and the remaining third in 2017. No never-treated group is included in this design.

The outcome variable is given by

$$Y_{it} = \alpha_i + \lambda_t + \varepsilon_{it} + \tau_{it},$$

with

$$\tau_{it} = 2 \cdot 1\{t \geq G_i\},$$

where $G_i \in \{1999,2008,2017\}$ denotes the treatment adoption year for unit i. All treatment groups experience the same constant treatment effect of 2 from the year of adoption onwards.

Because treatment effects are homogeneous across cohorts and constant over time, the TWFE estimator is expected to recover the correct average treatment effect in this design as well. However, the introduction of staggered timing foreshadows the complications that arise in subsequent designs, where heterogeneity in treatment effects leads TWFE to produce biased estimates

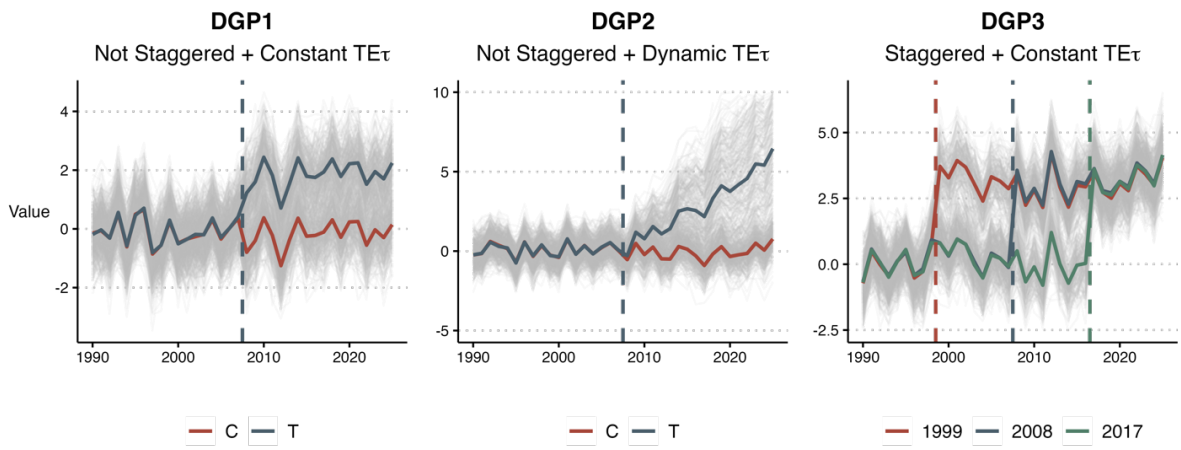

Figure 1: Outcome Path of DGP1-3

DGP 4: Staggered Adoption with Cohort-Specific Constant Effects

The fourth design builds on the staggered adoption structure of DGP 3, but introduces heterogeneity across cohorts. All units within a given treatment cohort experience a constant effect from the year of adoption onward, but the magnitude of the effect differs by adoption timing. Specifically, the treatment effect for unit i in cohort $g \in \{1999,2008,2017\}$ is

$$\tau_{it} \sim N(\delta_g \cdot 1\{t \geq g\},\ 0.2^2),$$

with $\delta_{1999} = 6$, $\delta_{2008} = 3$, and $\delta_{2017} = 1$.

This design reflects a scenario where early adopters experience large treatment gains, while later adopters benefit less. Because the average treatment effect varies systematically across cohorts, the TWFE estimator no longer corresponds to a meaningful weighted average of the cohort-specific effects. Some two-by-two comparisons involve later adopters being implicitly compared to earlier adopters who are already treated, which leads to biased estimates

DGP 5: Staggered Adoption with Homogeneous Dynamic Effects

The fifth design retains the staggered timing of DGP 3, but introduces dynamics in the treatment effect that evolve uniformly across cohorts. For each unit, the treatment

effect parameter δ_i is drawn from a normal distribution with mean 0.3 and variance 0.2^2:

$$\delta_i \sim N(0.3,\, 0.2^2).$$

The realized treatment effect for unit i in period t is then

$$\tau_{it} = \delta_i \cdot \max(0,\, t - G_i),$$

with variance proportional to the number of years since adoption:

$$\mathrm{Var}(\tau_{it}) = 0.2^2 \cdot \max(0,\, t - G_i).$$

This specification implies that treatment effects accumulate linearly over time at approximately 0.3 units per year after adoption with random variation across units. While the dynamic pattern is consistent across cohorts, the staggered timing again creates invalid comparisons in the TWFE framework, as some groups act as controls after they have already begun experiencing dynamic gains.

DGP 6: Staggered Adoption with Cohort-Specific Dynamic Effects

The sixth and most demanding design combines staggered adoption with both heterogeneity across cohorts and dynamic evolution of treatment effects. For each unit i in cohort g, the treatment effect parameter δ_i is drawn from

$$\delta_i \sim N(\delta_g, 0.2^2),$$

where the cohort-specific means are $\delta_{1999} = 0.5$, $\delta_{2008} = 0.3$, and $\delta_{2017} = 0.1$.

The unit–year treatment effect is then defined as

$$\tau_{it} = \delta_i \cdot \max(0,\, t - G_i),$$

with variance

$$\mathrm{Var}(\tau_{it}) = 0.2^2 \cdot \max(0,\, t - G_i).$$

This design represents a setting in which early adopters not only begin treatment earlier but also experience stronger annual gains, while later adopters experience smaller

dynamic effects. Such a structure reflects many empirical scenarios, such as policy interventions that are initially highly effective but yield diminishing returns for later adopters.

It also poses a severe challenge for the TWFE estimator, since the mixture of dynamic and heterogeneous effects exacerbates the weighting problems highlighted by the Goodman-Bacon decomposition.

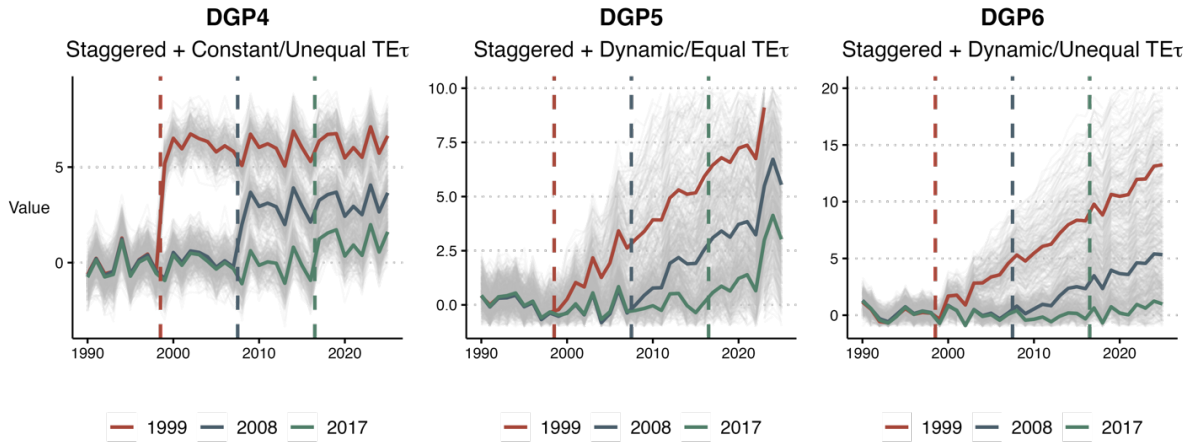

Figure 2: Outcome Path of DGP4-6

4.3 Estimation Strategy

The estimation strategy is based on repeated Monte Carlo simulations, which allow us to evaluate the finite-sample performances of different estimators under the six data-generating processes described in Section 4.2. For each DGP, a balanced panel dataset of 600 units observed over 36 years is generated. This process is repeated 500 times, and in each replications the estimators of interest are applied. The resulting estimates are then aggregated across replications to study their distribution, bias, and precisions to the true treatment effects defined in the DGP.

The first step in the analysis focuses exclusively on the two-way fixed effects (TWFE) estimator, implemented through a regression of the outcome variable on a treatment indicator, unit fixed effects, and time fixed effects. This setup corresponds to the standard specification commonly used in empirical applications. The TWFE results are

reported for all six DGPs. In particular, DGP1-3 serve as benchmark cases where TWFE is expected to recover the true effects, while DGP4-6 highlight scenarios where TWFE produces biased results due to treatment effect heterogeneity or dynamic patterns.

In the second step, attention shifts to the modern estimators of Callaway & Sant'Anna (2021) and Sun & Abraham (2021). These methods are applied only to DGP6, which represents the most demanding setting: staggered adoption combined with heterogeneous and dynamic treatment effects. By focusing on this design, the simulation highlights both the failure of TWFE and the ability of the newer estimators to recover valid estimates under precisely those conditions where TWFE breaks down.

Inference in the simulation study relies on clustered standard errors at the state level, reflecting the grouped structure of treatment assignment. For Callaway & Sant'Anna (2021), the group-time average treatment effects on treated are aggregated into dynamic effects using their proposed weighting scheme. For Sun & Abraham (2021), event-study coefficients are estimated in a way that excludes already-treated units from the control group, ensuring unbiased event-time estimates.

The presentation of results proceeds in two parts. First, graphical and distributional evidence is presented for TWFE under all six DGPs, showing when it succeeds and when it fails. Second, the performance of Callaway-Sant'Anna and Sun-Abraham is compared in DGP6, demonstrating how these methods provide reliable estimates in a setting where TWFE performs poorly.

5. Simulation Results for TWFE

5.1 Simulation Results for DGP1-3

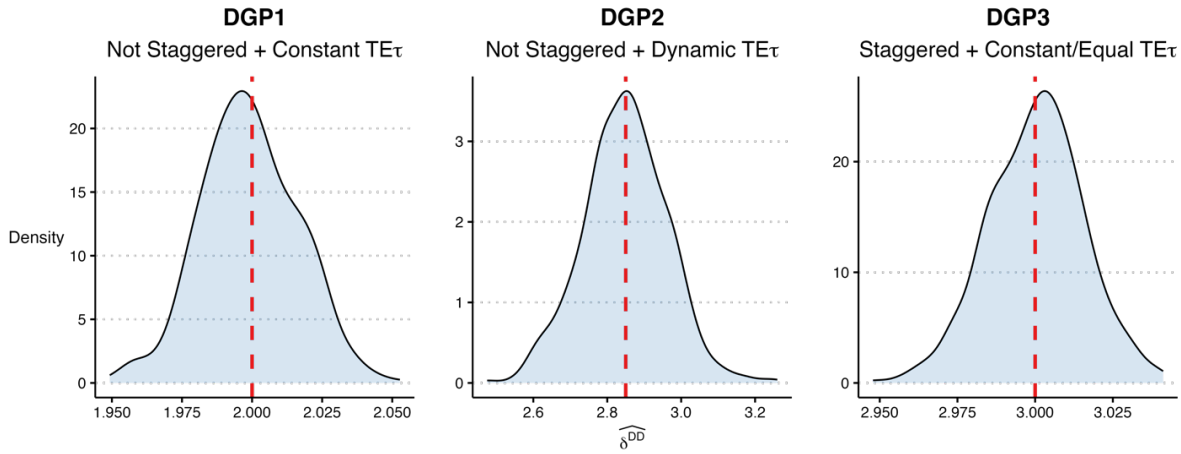

Figure 3: TWFE DiD estimates on DGP1-3

The first set of simulations examines scenarios where the assumptions underlying the TWFE estimator are satisfied. Figure 3 illustrates the distribution of the Monte Carlo estimates from 500 replications for DGP1-3, with the vertical red line marking the true average treatment effect on treated (ATT).

For DGP1, where half of the states adopt treatment in 2008 and treatment effects are constant across treated units, the TWFE estimator performs as expected. The distribution of estimates is centered closely around the true ATT, indicating negligible bias and confirming that TWFE recovers the correct effect under these conditions.

In DGP2, treatment effects are allowed to evolve dynamically over time following adoption in 2008. Despite the presence of dynamic treatment effects, the TWFE estimates remain centered on the true ATT. This result reflects the fact that all units adopt treatment at the same time, so no invalid comparisons arise between already treated and not-yet-treated groups.

DGP3 introduces staggered treatment adoption, with cohorts adopting in 1999, 2008, and 2017, but treatment effects remaining homogeneous across cohorts and constant

over time. Again, the TWFE estimator produces unbiased estimates, with the sample distribution aligned does not create bias, provide that treatment effects are homogeneous across groups and time.

Taken together, the results from DGP1-3 demonstrate that the TWFE estimator performs reliably in settings with a single adoption date or staggered adoption when treatment effects are constant across units and time. Bias only arises once heterogeneity across cohorts or over time is introduced, a case explored in DGP4-6.

5.2 Simulation Results for DGP4-6

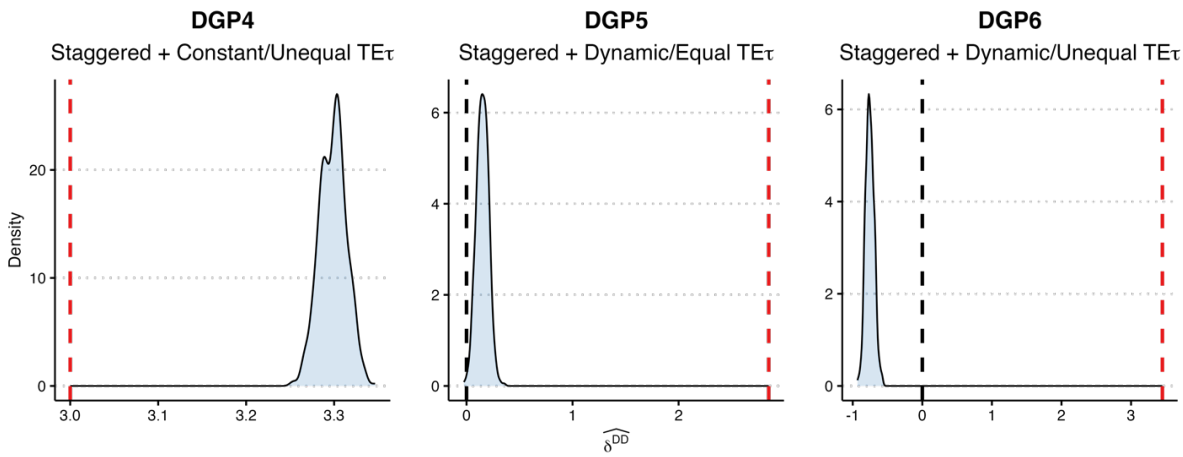

Figure 4: TWFE DiD estimates on DGP4-6

The second set of simulations introduces treatment effect heterogeneity across cohorts and over time. Figure 4 displays the distribution of TWFE estimates from 500 replications for DGP4-6, alongside the true average treatment effect on treated indicated by the red dotted line. In each case, the distribution of TWFE estimates is displaced from the true ATT, reflecting systematic bias.

In DGP4, where treatment effects are constant within cohorts but differ in magnitude across adoption groups, the TWFE estimator produces biased results. Although the average estimate remains positive, it fails substantially short of the true weighted ATT,

illustrating how TWFE mis-weights the contribution of groups with stronger and weaker effects.

DGP5 add dynamic treatment effects that accumulate over time but remain equal and constant speed across cohorts. Here too, the TWFE estimator fails to recover the true effect. The estimates are again biased but generally retain the correct positive sign, suggesting that while inference about the presence of an effect may be correct, the magnitude is misrepresented.

The most severe failure occurs in DGP6, where treatment effects are both dynamic and heterogeneous across cohorts. In this design, early adopters experience larger annual gains than later adopters. The TWFE estimates are not only biased but on average negative and statistically significant, even though the true effect for every cohort is positive in expectation. This reversal of sign highlights the severity of the weighting problem: the estimator combines group-specific effects in such a way that the overall estimate implies a treatment harm where in fact all groups benefit.

The intuition behind these biases can be traced to the decomposition of TWFE estimators developed by Goodman-Bacon (2021) as discussed in Section 2.3. TWFE estimates are weighted averages of all possible two-by-two DiD comparisons across cohorts. When treatment effects differ across groups or change dynamically over time, some of these comparisons use already-treated units as controls. This can assign negative or inappropriate weights to certain group-time effects, distorting the overall estimate. In DGP6, this re-weighting problem is so severe that it reverses the sign of the overall estimate, despite all cohorts experiencing positive treatment effects.

This findings from DGP4-6 confirm that the TWFE estimator cannot be relied upon when treatment effects are heterogeneous across cohorts or evolve dynamically over time. In particular, DGP6 illustrates how bias can become so severe that the estimator even reverses the sign of the treatment effect. This raises the question of whether alternative estimators, designed to address the shortcomings of TWFE, can perform

better in such settings. To answer this, Chapter 6 applies the methods of Callaway & Sant'Anna (2021) and Sun & Abraham (2021) to DGP6, the most challenging design in the simulation study.

6. Simulation Results for Callaway-Sant'Anna and Sun-Abraham methods

6.1 Applying Callaway-Sant'Anna and Sun-Abraham methods to DGP6

The results from Chapter 5 showed that the TWFE estimator fails dramatically in DGP6, where treatment effects are both dynamic and heterogeneous across adoption cohorts. In this setting, TWFE estimates are not only biased but misleading, suggesting negative treatment effects despite all groups benefiting in expectation. To address this problem, Chapter 6 turns modern Difference-in-Differences estimators specifically designed to handle staggered adoption and treatment effect heterogeneity.

Two alternative methods are considered: the estimator proposed by Callaway & Sant'Anna (2021) and the estimator developed by Sun & Abraham (2021). Both approaches avoid the pitfalls of TWFE by ensuring that already-treated units are not used as controls for later-treated groups. Instead, they rely on comparisons with either not-yet-treated or never-treated units, which preservers the validity of treatment-control contrasts even staggered adoption settings.

The Callaway-Sant'Anna estimator proceeds by first calculating group–time average treatment effects ($ATT_{g,t}$) for each cohort g and period t, using not–yet–treated units as the control group. These group–time effects are then aggregated into dynamic treatment effects, which trace how outcomes evolve over time relative to the adoption year.

Formally, the group–time average treatment effect is

$$ATT(g,t) = \mathbb{E}[\, Y_t(1) - Y_t(0) \mid G_g = 1 \,],$$

,

G_g as a binary variable equal to 1 if unit i first receives treatment in period g as mentioned Section 3.3. Estimation uses units that have not yet been treated by period t as the comparison group. The weight function used in this setting is dynamic treatment effect by event time.

Aggregating these effects yields dynamic treatment effect estimates that capture not only whether treatment effects are present, but also how they evolve in the years following adoption.

The Sun-Abraham estimator modifies the event-study regression framework commonly used in applied research. In their approach, the regression is reweighted so that the estimated coefficients correspond to comparisons between treated cohorts and not-yet-treated groups only. This correction ensures that the estimated event-study coefficients are unbiased, although it typically comes at the cost of larger standard errors due to the reduction in effective comparison groups.

Both methods are implemented on DGP6, which represents the most challenging environment in the simulation design. The results, presented in the next section, show how these estimators perform relative to the true treatment effect path and whether they succeed in overcoming the biases that plague TWFE.

6.2 Simulation Results for Callaway-Sant'Anna and Sun-Abraham methods.

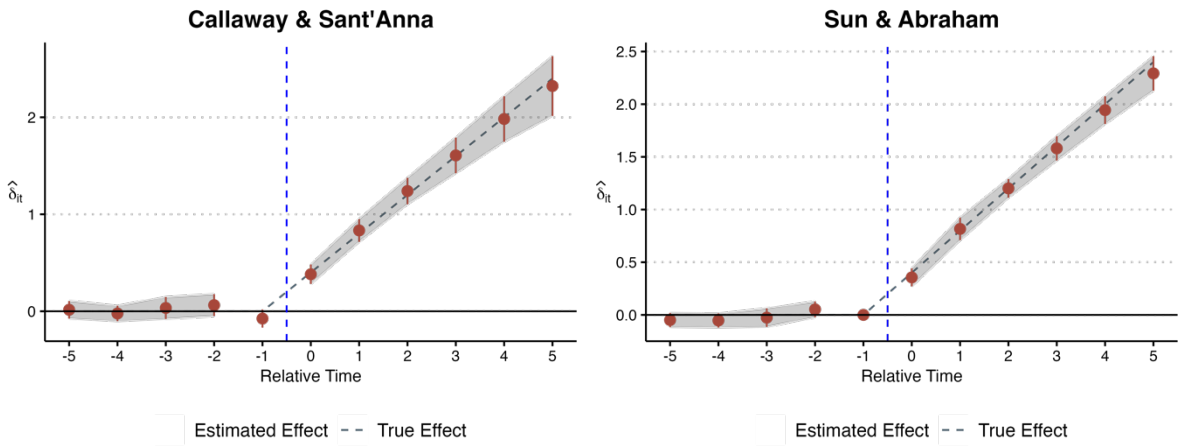

Figure 5: Callaway & Sant'Anna and Sun & Abraham methods with DGP6

Figure 5 displays the simulations results for the Callaway-Sant'Anna and Sun-Abraham estimators applied to DGP6. Recall that this design represents the most challenging setting, with staggered adoption combined with dynamic and heterogeneous treatment effects. In Chapter 5, the TWFE estimators was shown to fail severely in this case, producing biased and even negative estimates despite all cohorts experiencing positive treatment effects in expectation. The purpose of this section is therefore to assess whether the more recent estimators succeed where TWFE does not.

The results for the Callaway-Sant'Anna estimator show that the dynamic treatment effects are accurately recovered. The estimated effects closely follow the true treatment effect path, with point estimates lying near the dashed line that represent the simulated truth. Confidence intervals are wider in later periods, reflecting the smaller number of untreated comparison groups as more cohorts adopt treatment. Nonetheless, the estimator consistently detects positive effects and avoids the sign reversal observed under TWFE.

The Sun-Abraham estimator similarly performs well, producing event-study coefficients that align with the true treatment path. As expected, the estimator yields somewhat

larger standard errors than Callaway-Sant'Anna, because the correction excludes already-treated cohorts from the comparison group. Despite this, the estimates remain centered around the true effects, and the method successfully avoids the bias that distorted TWFE results.

Taken together, these results demonstrates that both Callaway & Sant'Anna (2021) and Sun & Abraham (2021) provide valid estimates under conditions where TWFE fails. By ensuring that already-treated units are not used as controls, both methods eliminate the negative weighting problem that plagued TWFE in DGP6. The trade-off is that precision is somewhat lower, especially for Sun-Abraham, but the key advantage is that the direction and magnitude of treatment effects are correctly identified.

7. Discussion and Conclusion

The simulation study presented in this thesis provides clear evidence on the performance of different Difference-in-Differences estimators across a range of multi-period time settings. The first set of designs (DGP1-3) demonstrated that the traditional two-way fixed effects estimator performs well when treatment effects are homogeneous and either constant or uniformly dynamic. In such cases, the estimates are centered on the true average treatment effect on treated (ATT), and staggered adoption by itself does not introduce bias. This result reassures researchers that TWFE remains a valid tool in settings where treatment adoption is simultaneous or where effects are stable across cohorts.

The picture changes dramatically once heterogeneity is introduced. In DGP4-6, the estimator consistently fails to recover the true ATT. The most severe case arises in DGP6, where treatment effects are both dynamic and differ across adoption cohorts. Here, the TWFE estimates are not just biased but misleading, often suggesting negative treatment effects even though all groups benefit in expectation. This reversal of sign illustrates the weighting problem described by Goodman-Bacon (2021): TWFE

aggregates two-by-two comparisons across cohorts, and when some of these comparisons involve already-treated units serving as controls, the resulting weights can distort the overall estimate to the point of reversing its direction.

The application of Callaway & Sant'Anna (2021) and Sun & Abraham (2021) estimators to DGP6 shows that these modern approaches provide a solution to the problems that undermine TWFE. By ensuring that already-treated cohorts are excluded from the control group, both methods recover treatment effect paths that closely match the true simulated values. The Callaway & Sant'Anna (2021) estimator produces relatively precise estimates, while the Sun & Abraham(2021) approach yields wider confidence intervals but remains unbiased. Together, these results highlight a clear trade-off: modern estimators may sacrifice some efficiency, but they deliver validity, which is ultimately more important for credible empirical inference.

These findings have important implications for applied research. Many policy evaluations involve interventions that are rolled out at different times across states, firms, or regions. In such settings, treatment effects are rarely homogeneous, and relying on TWFE can therefore lead to misleading conclusions. The simulations here show that even the sign of the estimated effect can be wrong, with potentially serious consequences for policy interpretation. By contrast, methods such as Callaway & Sant'Anna (2021) and Sun & Abraham(2021) are better suited to these environments and should be preferred whenever staggered adoption and treatment effect heterogeneity are present.

The thesis also provides guidance for researchers. TWFE is still a reasonable choice when treatment is adopted in a single period across all units, or when staggered adoption occurs but effects are constant across groups. In all other cases, particularly those with dynamic or heterogeneous treatment effects, modern alternatives are essential. A simple rule of thumb emerges: the greater the heterogeneity in treatment

timing and effects, the less reliable TWFE becomes, and the stronger the case for adopting estimators such as Callaway & Sant'Anna (2021) or Sun & Abraham, (2021).

At the same time, the simulation study in subject to limitations. The designs abstract from several complications that frequently arise in real data, including selective treatment adoption, attrition, the presence of covariates, and possible violations of the parallel trends assumption. The number of units and time periods was fixed, and only certain forms of heterogeneity were modeled. As a result, the conclusions should be viewed as illustrative rather than exhaustive.

These limitations point to natural directions for future research. One extension would be to introduce observed covariates into simulation and evaluate how modern estimators perform when adjustment is required. Another would be to allow for selective adoption, where units with certain characteristics are more likely to adopt treatment earlier, reflecting realistic policy settings. Finally, future work could investigate the robustness of Callaway & Sant'Anna (2021) and Sun & Abraham (2021) under departures from the parallel trends assumption. Pursuing these avenues would bring simulation designs closer to the conditions faced in empirical applications and strengthen the practical guidance available to applied researchers.

In conclusion, this thesis has shown that Difference-in-Differences methods in multiple time periods used to rely on TWFE. While TWFE remains appropriate in limited circumstances, it cannot be relied upon in the presence of staggered adoption and heterogeneous treatment effects. Under such conditions, modern estimators such as Callaway & Sant'Anna (2021) and Sun & Abraham (2021) provide and credible alternatives. Although they involve a trade-off in terms of precision, their ability to avoid the severe biases documented in the simulations makes them essential tools for applied research.

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I hereby confirm that the work presented has been performed and interpreted solely by myself except for where I explicitly identified the contrary. I assure that this work has not been presented in any other form for the fulfillment of any other degree or qualification. Ideas taken from other works in letter and in spirit are identified in every single case.

Bonn, 01/09/2025_____

Duc Thinh Phung